

Quiz (Carolina Reaper)

- Q1.** What is the purpose of a scatter plot?
(A) To show the relationship between two variables
(B) To display a single data point
(C) To create a line graph
(D) To make predictions without data
(E) To draw a circle
- Q2.** If x increases as y increases, what type of correlation is present?
(A) Negative
(B) Positive
(C) None
(D) Zero
(E) Inverse
- Q3.** Which of the following is the equation of a line of best fit?
(A) $y = 2x + 3$
(B) $y = x^2 + 2$
(C) $x = 2y + 1$
(D) $y = \frac{1}{x}$
(E) $x = y^2 - 1$
- Q4.** What is used to determine the line of best fit?
(A) The y -intercept
(B) The slope of the line
(C) The correlation coefficient
(D) The x -intercept
(E) All of the above
- Q5.** What does a correlation coefficient of 1 indicate?
(A) No correlation
(B) Negative correlation
(C) Positive correlation
(D) A perfect circle
(E) A vertical line
- Q6.** A line of best fit can be used for what purpose?
(A) To make predictions
(B) To determine the y -intercept
(C) To graph a circle
(D) To find the correlation coefficient
(E) To draw a bar graph
- Q7.** If x decreases as y decreases, what type of correlation is present?
(A) Negative
(B) Positive
(C) None
(D) Zero
(E) Inverse
- Q8.** What does the slope of the line of best fit represent?
(A) The change in y for a 1-unit change in x
(B) The change in x for a 1-unit change in y
(C) The y -intercept
(D) The x -intercept
(E) The correlation coefficient

Open Ended Questions (FRQ)

- Q9.** Given the data points (2, 3), (4, 5), (6, 7), and (8, 9), create a scatter plot and draw the line of best fit. Use the line to make a prediction for y when $x = 10$.
- Q10.** Describe how the line of best fit is used to make predictions based on data trends. Provide an example of a real-world scenario where the line of best fit is used.

Chef's Tips

- Tip 1: Ensure students use correct units and labels on their graphs
- Tip 2: Have students check their calculations for accuracy
- Tip 3: Encourage students to use real-world examples to illustrate their answers